

Influence Maximization When Communication Is the Bottleneck

From centralized submodular greedy to CELF acceleration and local candidate sharing

Yixin Chen Bichi Zhang Yaoqin Ye

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Research question: how much influence quality can we keep when expensive centralized seed selection is replaced by cached greedy evaluation or communication-limited local candidates?

1. Optimization Target

We choose k initial seeds in a network and maximize expected diffusion coverage:

$$\max_{S \subseteq V, |S| \leq k} \sigma(S).$$

Here $\sigma(S)$ is estimated by Monte Carlo simulation under the Independent Cascade model.

2. Diffusion Model

► Newly activated nodes get one chance to activate each inactive out-neighbor.

► Each edge attempt succeeds independently with probability

p_{uv} .

► We use the weighted-cascade rule:

$$p_{uv} = \frac{1}{\text{indegree}(v)}.$$

3. Why Greedy Is Principled

IC influence spread is monotone and submodular:

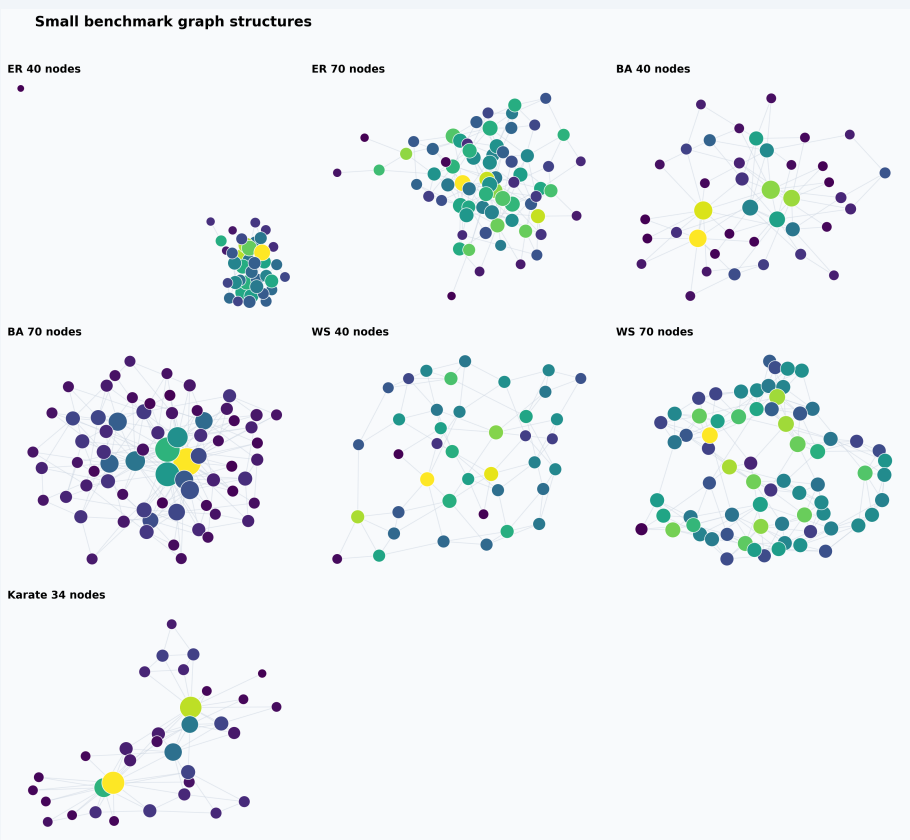
$$\sigma(A \cup \{v\}) - \sigma(A) \geq \sigma(B \cup \{v\}) - \sigma(B).$$

► Exact influence maximization is NP-hard.

► Greedy has a $(1 - 1/e)$ guarantee for the exact objective.

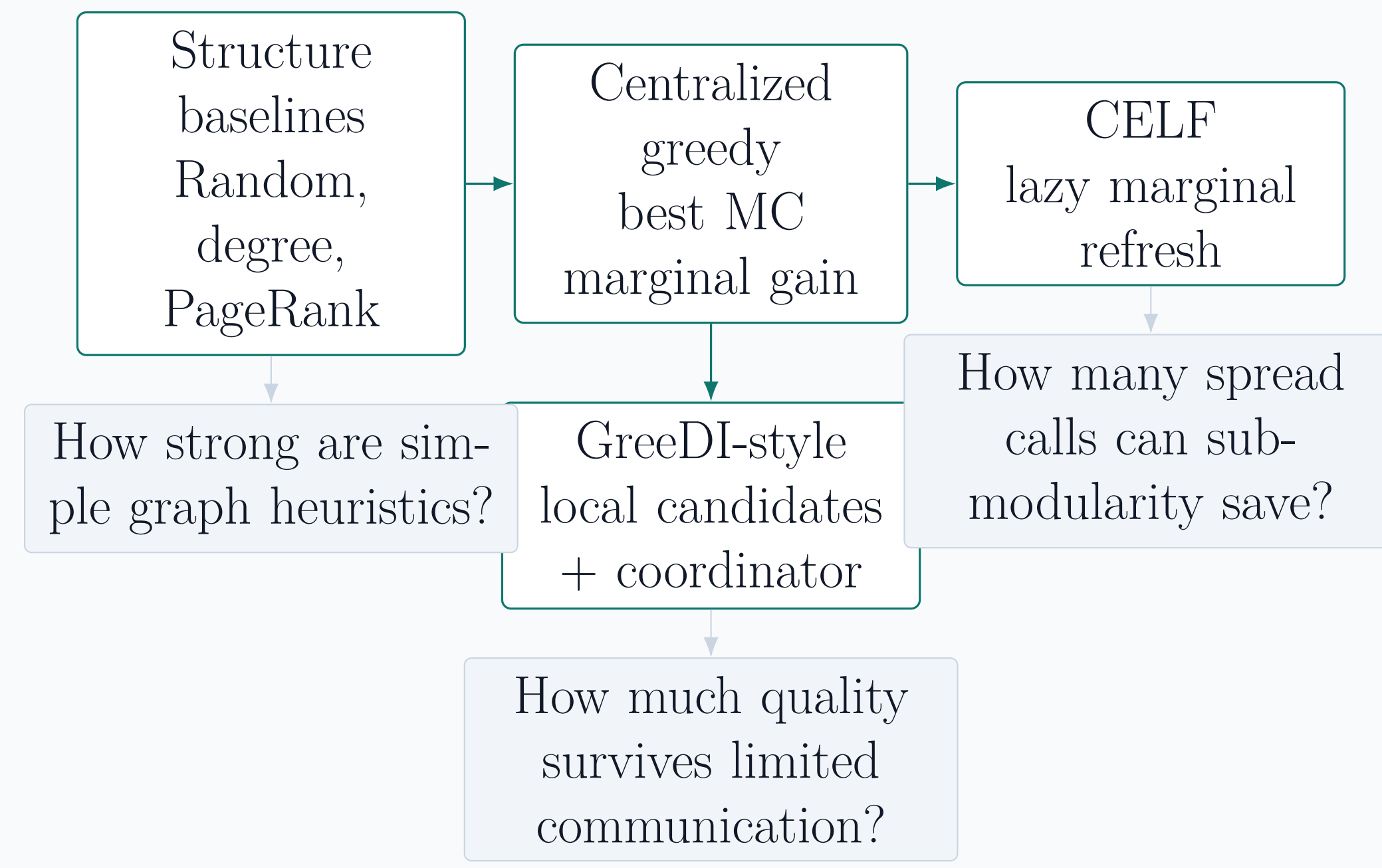
► The bottleneck is not the rule itself; it is repeated spread estimation.

Benchmark Graphs

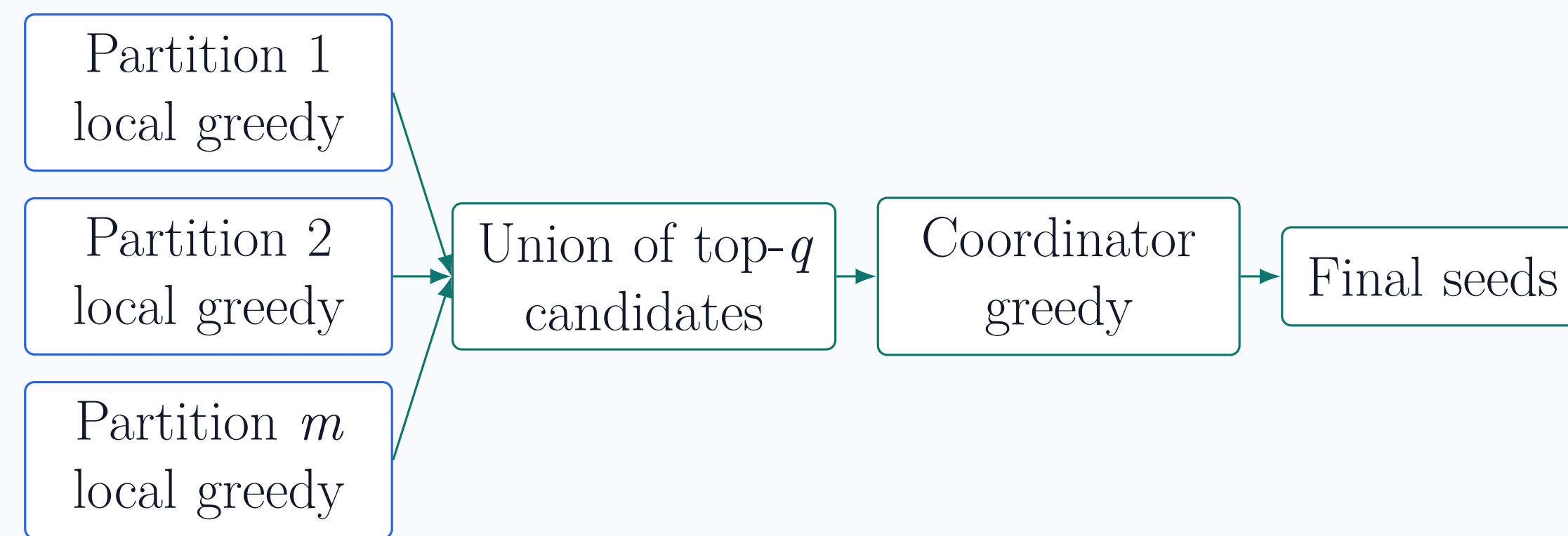


Random, hub-heavy, small-world, and Karate Club graph structures expose different seed-selection behavior.

Algorithmic Story



Distributed Candidate Sharing



Each worker sends at most q candidate nodes. The coordinator searches only the merged candidate pool.

Experimental Protocol

Graph families	Erdos-Renyi, Barabasi-Albert, Watts-Strogatz, Karate
Large sizes	$n \in \{50, 100, 200\}$, plus Karate
Seed budget	$k = 10$ for synthetic large graphs, $k = 5$ for Karate
Distributed sweep	$m \in \{2, 4, 8\}$, $q \in \{k, 2k, 5k\}$
Fair evaluation	final seed sets re-estimated with a common seed

74.1% fewer evaluations
CEL F vs. greedy

Large-scale average: 266.2 vs. 1026.5 spread evaluations.

83.5% lower runtime
CEL F vs. greedy

Large-scale average: 0.77s vs. 4.67s wall-clock time.

Large-Scale Evidence

Method	Spread	Ratio	Time	Eval.
Random	31.59	0.791	0.00s	1.0
W. Degree	43.71	1.066	0.01s	1.0
PageRank	35.19	0.893	0.00s	1.0
Greedy	40.43	1.000	4.67s	1026.5
CEL F	40.05	0.988	0.77s	266.2

- CEL F nearly preserves greedy spread but sharply cuts spread calls.
- Weighted degree is not a weak baseline under weighted-cascade probabilities.
- Distributed quality remains competitive, but communication cost rises quickly.

Case-Level Results



Communication Tradeoff

	Budget	Ratio	Time	Eval.	Cand.
$q = k$	0.980	1.58s	1242.4	41.1	
$q = 2k$	0.965	3.36s	2037.2	66.9	
$q = 5k$	0.955	6.48s	3044.4	98.4	

Bigger candidate budgets transmit more nodes and trigger more evaluations. In this noisy course-scale setting, quality does not improve monotonically with q .

Takeaways

- Submodularity gives the project a principled greedy baseline.
- CEL F is the clearest algorithmic improvement: it keeps the target but removes redundant evaluations.
- Communication-limited candidate sharing changes the cost model: less global search, more sensitivity to partitions and Monte Carlo noise.

Implementation. IC simulation, greedy selection, CEL F, and distributed candidate sharing are implemented as separate modules.

Run command. `python src/experiments.py --scale large --seed 7 --simulations 40`

Scope. This is a course-scale reproduction of algorithmic mechanisms, not a full IMM/RIS or production distributed system.